Year 1 AS Statistics and Mechanics

Chapter 8

Modelling in mechanics

Practice Booklet

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8.1 Constructing a model

Practice Book, printed page 52.

1. The motion of a football after it is kicked by a player can be modelled using the equation $h = 0.6x - 0.06x^2$, where h m is the height of the football above the ground and x m is the horizontal distance travelled.

a. Find the height of the football when it is:

i. struck

ii. at a horizontal distance of 3 m

iii. at a horizontal distance of 6 m.

b. Use the model to predict the height of the football when it is 15 m from the player.

c. Comment on the validity of this prediction.

Hint

When the football is struck, $x = 0$.

Hint

To answer part **c**, look at the context of the question to determine whether your answer is valid. In this case, h represents the height of the football above the ground so the model is only valid for $h \geq 0$.

2. A diver jumps from a diving board into a swimming pool with a depth of 5 m. The height of the diver above the water level, h m, after t seconds can be modelled using the equation $h = -5t^2 + 3t + 8$.

a. Write down the height of the diving board above the water.

b. Find the height of the diver above the water when $t = 1$.

c. Use the model to predict the height of the diver after 3 seconds.

d. Comment on the validity of this prediction.

3. When Rashid takes a throw in the shot put, the motion of the shot can be modelled using the equation $h = 0.8x - 0.05x^2$, where h m is the height of the shot above Rashid's chest (the point from which it's thrown) and x m is the horizontal distance travelled.

a. Find the height of the shot above Rashid's chest when:

i. $x = 2$

ii. $x = 5$

iii. $x = 10$

b. Use the model to predict the height of the shot above Rashid's chest when $x = 20$. Comment on the validity of this prediction.

4. P A car decelerates from 60 mph to 20 mph in 8 seconds. A quadratic equation of the form $d = 26.8t - kt^2$ can be used to model the distance travelled by the car, where d metres is the distance travelled in time t seconds.

a. Given that when $t = 2$ the distance travelled by the car is 49.1 m, use the model to find the distance travelled when the car reaches 20 mph.

b. Write down the range of values of t for which the model is valid.

Hint

Substitute the values given in the question into the model and solve the corresponding equation to find k.

5. P The model for the motion of a football given in question 1 is only valid when h is positive. Find the range of values of x for which the model is valid.

6. P The model for the height of the diver above the water given in question 2 is only valid from the time the diver dives until the time the diver enters the water. Find the range of values of t for which the model is valid.

7. P The model for the motion of Rashid's shot given in question 3 is only valid when h is positive.

a. Find the range of values of x for which the model is valid.

b. Work out the maximum height of Rashid's shot above his chest.

When Sam takes a throw in the shot put, the motion of the shot can be modelled using the equation $h = 0.6x - 0.04x^2$, where h m is the height of the shot above his chest and x m is the horizontal distance travelled. The player that throws the shot furthest wins the competition.

c. Use the models given to determine who will win the contest, and find the distance by which they win. Assume the shot finishes at the same height as it started in each case.

8. E/P The motion of a netball as it leaves a player's hand and passes through the net is modelled using the equation $h = 1.8 + 1.2x - 0.2x^2$, where h m is the height of the netball above the ground and x m is the horizontal distance travelled.

a. Write down the height of the ball when it leaves the player's hand. (1 mark)

b. Find the two values of x for which the netball is exactly 3.4 m above the ground. (2 marks)

The model is only valid for $0 \leq x \leq k$, where k m is the horizontal distance of the net from the player. Given that the height of the net is 2.8 m and that the netball passes through the net as it is travelling downwards:

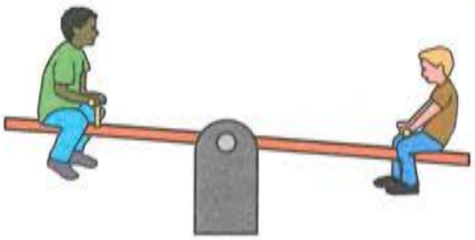
c. find the range of values of x for which the model is valid (1 mark)

d. work out the maximum height of the netball. (2 marks)

8.2 Modelling assumptions

Practice Book, printed page 53.

1. Two children of different masses are sitting on a see-saw that rests on a support.



State the effect on any calculations made using this model based on the following assumptions:

a. the children are modelled as particles

b. the see-saw is modelled as a uniform rod

c. the support is modelled as a smooth peg

d. air resistance is negligible

Hint

Modelling assumptions simplify a problem. For example, by modelling a child as a particle you assume that the child's weight acts at a single point.

2. A mechanical engineer constructs a mathematical model to predict the behaviour of a car on a particular racetrack. The engineer models the racetrack as smooth.

a. Describe the effects of this modelling assumption.

b. Comment on its validity in this situation.

Hint

Modelling assumptions can affect the validity of a model. Think about whether modelling the racetrack as smooth is a reasonable assumption.

3. A bungee jumper wants to model her descent. She models the bungee cord as an inextensible string. Explain why this modelling assumption is unlikely to be suitable for this situation.

4. Make a list of the assumptions you might make to model a car and trailer connected by a tow-bar.

5. P A small stone is dropped from a height and falls vertically to the ground.

a. Make a list of the assumptions you might make to create a simple model.

b. A feather is dropped from the same height as the stone. Describe the differences between the motion of the stone and the motion of the feather:

i. using the same model for the feather and the stone

ii. in real life.

c. With reference to your answer to part **b**, suggest a possible refinement to your model.

8.3 Quantities and units

Practice Book, printed page 54.

1. Convert to base SI units:

a. $45\,\text{km}\,\text{h}^{-1}$

b. 1800 g per minute

c. $7\,\text{g}\,\text{cm}^{-3}$

d. $540\,\text{g}\,\text{m}^{-3}$

e. $1.2 \times 10^{-4}\,\text{kg}\,\text{cm}^{-3}$

f. $4.1 \times 10^{5}\,\text{g}\,\text{cm}^{-3}$

Hint

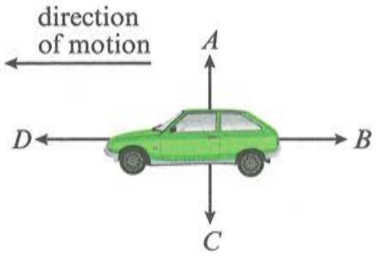
Kilograms (kg), metres (m) and seconds (s) are all base SI units.

2. Write down the names of the forces shown in each of these diagrams.

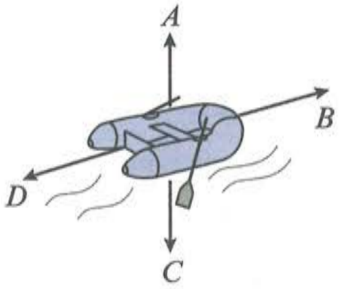
a. A parachute jumper descending to the ground after opening the parachute.



b. A car driving along a level road.



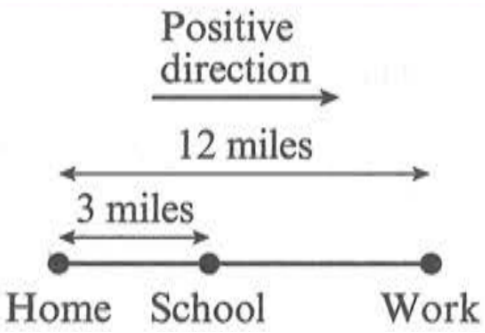
c. An inflatable dinghy being towed along a lake by a boat.



8.4 Working with vectors

Practice Book, printed page 55.

1. Julia drives from her home along a straight road to work at a constant speed of 42 mph. On her way home from work she stops at her daughter's school to pick her up. She drives from work to the school at a constant speed of 28 mph. She drives from the school to her home at a constant speed of 34 mph. Julia's home, work and the school all lie on a straight line, as shown in the diagram.



Taking the positive direction as shown in the diagram, state Julia's:

a. velocity on the journey from her home to work

b. displacement from her home when she reaches work

c. velocity on the journey from work to the school

d. displacement from work when she reaches the school

e. velocity on the journey from the school to her home.

Hint

You need to choose a positive direction when considering motion in a straight line. When an object is moving in the opposite direction to the positive direction its velocity is negative.

2. Match each scenario **i**–**iv** with the motion of each object **a**–**d**.

i. A ball falls to the ground and its speed increases.

ii. A ball is thrown upwards and its speed decreases.

iii. A rocket is launched upwards and its speed increases.

iv. A helicopter lowers to the ground and its speed decreases.

	a	**b**	**c**	**d**
Velocity	+ve	+ve	−ve	−ve
Acceleration	+ve	−ve	−ve	+ve



Hint

When the sign of the acceleration and the sign of the velocity are different, the acceleration is opposing the direction of motion, so the particle is slowing down.

3. The velocity of a particle is given by $\mathbf{v} = 8\mathbf{i} + 12\mathbf{j}\,\mathrm{m\,s^{-1}}$. Find:

a. the speed of the particle

b. the angle the direction of motion of the particle makes with the unit vector $\mathbf{i}$.

Hint

Draw a diagram. The speed of the particle is the magnitude of the vector $\mathbf{v}$. You can use trigonometry to find the direction of $\mathbf{v}$.

4. The acceleration of a car is given by $\mathbf{a} = -7\mathbf{i} + 5\mathbf{j}\,\text{m}\,\text{s}^{-2}$. Find:

a. the magnitude of the acceleration of the car

b. the angle the direction of the acceleration vector of the car makes with the unit vector $\mathbf{j}$.

5. E/P A car drives from A to B and then from B to C.
The displacement from A to B is $10\mathbf{i} + 5\mathbf{j}$ km.
The displacement from B to C is $16\mathbf{i} - 13\mathbf{j}$ km.

a. Find the magnitude of the displacement from A to C. (3 marks)

b. Find the angle $\overrightarrow{AC}$ makes with the unit vector $\mathbf{i}$. (2 marks)

6. E/P A man jogs in a straight line from P to Q and then in a straight line from Q to R.
The displacement from P to Q is $240\mathbf{i} - 600\mathbf{j}$ m.
The displacement from Q to R is $450\mathbf{i} + 350\mathbf{j}$ m.

a. Find the magnitude of the displacement from P to R. (3 marks)

b. Find the total distance, correct to the nearest metre, that the man has jogged in getting from P to R. (3 marks)

c. Find the angle that $\overrightarrow{PR}$ makes with the unit vector $\mathbf{j}$. (2 marks)

Problem solving: Set A

Bronze

The motion of a ball can be modelled using the equation $h = 1.4 + 0.5x - 0.1x^2$, where h m is the height of the ball above the ground and x m is the horizontal distance travelled.

a. Find the height of the ball when:

i. $x = 2$

ii. $x = 4$

iii. $x = 6$

(3 marks)

b. Use the model to predict the height of the ball when $x = 10$. Comment on the validity of this prediction. (2 marks)

Silver

The motion of a volleyball after it is served by a player is modelled using the equation $h = 1.2 + x - 0.2x^2$, where h m is the height of the volleyball above the ground and x m is the horizontal distance travelled.

a. Write down the height of the ball when it leaves the player's hand. (1 mark)

The model for the motion of a volleyball is only valid when h is positive.

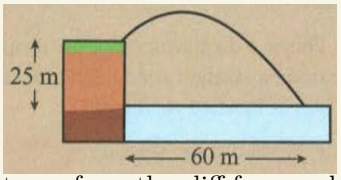
b. Find the range of values of x for which the model is valid. (2 marks)

c. A player stands k m from the net when the volleyball is served. The volleyball net is 2 m high. Given that the volleyball clears the net, find the range of possible values of k. (3 marks)

d. Find the maximum height of the volleyball according to this model. (2 marks)

Gold

A golf ball is struck from the top of a vertical cliff 25 m above water level. It enters the water at a horizontal distance of 60 m from the cliff face.
The motion of the golf ball is modelled by the equation $h = px - 0.02x^2 + q$, $0 \leq x \leq 60$,



where h m is the height of the golf ball above the water, x m is the horizontal distance from the cliff face, and p and q are constants.
Find the maximum height of the golf ball above the water. (4 marks)

Problem solving: Set B

In these questions, the unit vector **j** is directed due north and the unit vector **i** is directed due east.

Bronze

The velocity of a particle is given by $\mathbf{v} = 5\mathbf{i} + 6\mathbf{j}\,\text{m s}^{-1}$. Find:

a. the speed of the particle (2 marks)

b. the angle the direction of motion of the particle makes with the unit vector **i**. (2 marks)

Silver

An aeroplane flies from airport A to make three deliveries at airports B, C and D, in that order.
The displacement from A to B is $120\mathbf{i} - 200\mathbf{j}$ km.
The displacement from B to C is $150\mathbf{i} + 180\mathbf{j}$ km.
The displacement from C to D is $-250\mathbf{i} + 80\mathbf{j}$ km.
The aeroplane has enough fuel to fly 800 km, and flies in a straight line between each airport.

a. Does the aeroplane have enough fuel make the three deliveries and return to airport A? Explain your answer fully. (4 marks)

b. Find the bearing from D to A. Give your answer as a three-figure bearing. (3 marks)

Gold

A hiker walks in a straight line from P to Q and then in a straight line from Q to R.
The displacement from P to Q is $900\mathbf{i} - 400\mathbf{j}$ m.
The displacement from Q to R is $-500\mathbf{i} + a\mathbf{j}$ m.
The magnitude of the displacement from P to R is 412.3 m.
The total distance the hiker travels from P to R is 1568 m.
Find the value of a to the nearest metre, showing your working clearly. (6 marks)

Answers

8.1 Constructing a model

1.

a.

i. 0 m

ii. 1.26 m

iii. 1.44 m

b. -4.5 m

c. Model is not valid when $x = 15$ as height would be 4.5 m below ground level.

2.

a. 8 m

b. 6 m

c. -28 m

d. The model is not valid when $t = 3$. After the diver enters the water, other forces then act on the diver and so the model ceases to be valid.

3.

a.

i. 1.4 m

ii. 2.75 m

iii. 3 m

b. $h = -4$ m; so the model is not valid when $x = 20$ as the height would be below ground level.

4.

a. 142.4 m

b. The model is valid for $0 \leq t \leq 8$

5. $0 \leq x \leq 10$

6. $0 \leq t \leq 1.6$

7.

a. $0 \leq x \leq 16$

b. 3.2 m

c. Rashid throws the shot 16 m, Sam throws the shot 15 m. Rashid wins by 1 m.

8.

a. 1.8 m

b. $x = 2$ and $x = 4$

c. $0 \leq x \leq 5$

d. 3.6 m

8.2 Modelling assumptions

1.

a. Ignores the rotational effect of any external forces that are acting on the see-saw, and the effects of air resistance.

b. Ignores the width of the see-saw and assumes that the weight acts at the midpoint of the rod.

c. This presumes the support is dimensionless and fixed, and ignores any frictional effects on the rod.

d. Ignores the frictional effects on the children due to air resistance.

2.

a. Ignores the friction between the racetrack and the car.

b. The assumption is not valid; the racetrack will exert frictional forces on the car.

3. The bungee cord will stretch under the weight of the jumper.

4. Model the car and trailer as particles connected by a light rod.
Ignore the effects of air resistance.

5.

a. Model the stone as a particle, ignore the effects of air resistance.

b.

i. The motion will be identical.

ii. The feather will take longer to fall.

c. Include air resistance in the model, for example.

8.3 Quantities and units

1.

a. $12.5\,\mathrm{m\,s^{-1}}$

b. $0.03\,\mathrm{kg\,s^{-1}}$

c. $7000\,\mathrm{kg\,m^{-3}}$

d. $0.54\,\mathrm{kg\,m^{-3}}$

e. $120\,\mathrm{kg\,m^{-3}}$

f. $4.1 \times 10^{8}\,\mathrm{kg\,m^{-3}}$

2.

a. A: Air resistance, B: Weight.

b. A: Normal reaction, B: Friction or air resistance, C: Weight, D: Forward thrust.

c. A: Buoyancy, B: Tension, C: Weight, D: Water resistance or drag.

8.4 Working with vectors

1.

a. 42 mph

b. 12 miles

c. -28 mph

d. -9 miles

e. -34 mph

2.

a. iii
b. ii
c. i
d. iv

3.

a. $14.4\,\mathrm{m\,s^{-1}}$ (3 s.f.)
b. $56.3°$ (3 s.f.)

4.

a. $8.60\,\mathrm{m\,s^{-2}}$ (3 s.f.)
b. $54.5°$ (3 s.f.)

5.

a. 27.2 km (3 s.f.)
b. $17.1°$ (3 s.f.)

6.

a. 733.9 m (4 s.f.)
b. 1216 m
c. $109.9°$ (4 s.f.)

Problem solving: Set A

Bronze

a.

i. 2 m
ii. 1.8 m
iii. 0.8 m

b. $h = -3.6$ m, so the model is not valid when $x = 10$ as the model predicts the ball would be 3.6 m below the ground.

Silver

a. 1.2 m
b. $0 \leq x \leq 6$
c. $1 < k < 4$
d. 2.45 m

Gold 32.7 m (3 s.f.)

Problem solving: Set B

Bronze

a. $7.81\,\mathrm{m\,s^{-1}}$ (3 s.f.)
b. $50.2°$ (3 s.f.)

Silver

a. Yes.

$$\begin{aligned}\text{Distance travelled} &= 233.24 + 234.31 + 262.49 + 63.25 \\ &= 793.29 \text{ km} < 800 \text{ km}\end{aligned}$$

b. 198°

Gold $a = 300$